

Session – 2 :

Java Sort :

```
import java.io.*;

import java.util.*;

interface PerformOperation {

    boolean check(int a);

}

class MyMath {

    public static boolean checker(PerformOperation p, int num) {

        return p.check(num);

    }

    PerformOperation isOdd() {

        return n -> n % 2 != 0;

    }

    PerformOperation isPrime() {

        return n -> {

            if (n < 2) return false;

            for (int i = 2; i * i <= n; i++) {

                if (n % i == 0)

                    return false;

            }

            return true;

        };

    }

    PerformOperation isPalindrome() {

        return n -> {

            int temp = n, rev = 0;

            while (temp > 0) {

                rev = rev * 10 + temp % 10;

                temp /= 10;

            }

        }

    }

}
```

```

        return rev == n;

    };

}

}

public class Solution {

    public static void main(String[] args) throws IOException {

        MyMath ob = new MyMath();

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

        int T = Integer.parseInt(br.readLine());

        PerformOperation op;

        boolean ret = false;

        String ans = null;

        while (T--> 0) {

            String s = br.readLine().trim();

            StringTokenizer st = new StringTokenizer(s);

            int ch = Integer.parseInt(st.nextToken());

            int num = Integer.parseInt(st.nextToken());

            if (ch == 1) {

                op = ob.isOdd();

                ret = ob.checker(op, num);

                ans = (ret) ? "ODD" : "EVEN";

            } else if (ch == 2) {

                op = ob.isPrime();

                ret = ob.checker(op, num);

                ans = (ret) ? "PRIME" : "COMPOSITE";

            } else if (ch == 3) {

                op = ob.isPalindrome();

                ret = ob.checker(op, num);

                ans = (ret) ? "PALINDROME" : "NOT PALINDROME";

            }

            System.out.println(ans);

```

```
}  
}  
}
```

Output:

✓ Test case 0

✓ Test case 1

✓ Test case 2

✓ Test case 3

✓ Test case 4

✓ Test case 5

✓ Test case 6

1 5

2 33 Rumpa 3.68

3 85 Ashis 3.85

4 56 Samiha 3.75

5 19 Samara 3.75

6 22 Fahim 3.76

Expected Output

1 Ashis

2 Fahim

3 Samara

4 Samiha

5 Rumpa

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